# LCM & HCF

## Basic to Practical Level

### Quick Revision & Calculation Speed Guide

**Purpose of These Notes**

These notes are designed to revise the concepts of Factors, Multiples, HCF, LCM, Prime Factorization, Divisibility and related calculation techniques quickly.
The focus is on understanding concepts and improving calculation speed rather than advanced competitive-exam tricks.

Mathematics Revision Notes

Basic → Intermediate → Practical

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# 1 Number Basics

Before learning HCF and LCM, it is important to understand a few basic ideas about numbers.

## 1.1 Natural Numbers

Natural numbers are the counting numbers.

$$1, 2, 3, 4, 5, 6, \ldots$$

Depending on the convention being used, some books include 0 in natural numbers. For these notes, we will use:

$$\mathbb{N} = \{1, 2, 3, \ldots\}$$

## 1.2 Whole Numbers

Whole numbers include zero and all positive integers.

$$0, 1, 2, 3, 4, 5, \ldots$$

## 1.3 Integers

Integers include positive numbers, negative numbers and zero.

$$\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$$

## 1.4 Even Numbers

A number divisible by 2 is called an even number.

Examples:

$$2, 4, 6, 8, 10, 12, 14, \ldots$$

Every even number can be written as:

$$2n$$

where $n$ is an integer.

### 1.5 Odd Numbers

A number that is not divisible by 2 is called an odd number.

Examples:

$$1, 3, 5, 7, 9, 11, \ldots$$

Every odd number can be written as:

$$2n + 1$$

### 1.6 Prime Numbers

A prime number has exactly two positive factors:

$$1 \quad \text{and itself}$$

Examples:

$$2, 3, 5, 7, 11, 13, 17, 19, \ldots$$

**Remember**

The number 1 is neither prime nor composite.
Also, 2 is the only even prime number.

### 1.7 Composite Numbers

A composite number has more than two positive factors.

Examples:

$$4, 6, 8, 9, 10, 12, 15, 16, \ldots$$

For example:

$$12 = 1 \times 12 = 2 \times 6 = 3 \times 4$$

Therefore, 12 has more than two factors and is composite.

## 2 Factors

**Definition**

A factor of a number is a number that divides it exactly without leaving a remainder.

For example, consider 12.

$$12 \div 1 = 12$$

$$12 \div 2 = 6$$

$$12 \div 3 = 4$$

$$12 \div 4 = 3$$

$$12 \div 6 = 2$$

$$12 \div 12 = 1$$

Therefore, the factors of 12 are:

$$\boxed{1, 2, 3, 4, 6, 12}$$

## 2.1 Factor Pairs

Factors can be grouped into pairs whose product is the original number.
For 12:

$$1 \times 12$$

$$2 \times 6$$

$$3 \times 4$$

Therefore, the factor pairs of 12 are:

$$(1, 12), \quad (2, 6), \quad (3, 4)$$

## 2.2 Important Properties of Factors

For a positive integer $n$:

- 1 is always a factor.
- $n$ itself is always a factor.

- A factor cannot be greater than the number itself.
- Every factor divides the number exactly.

**Example**

Find the factors of 18.

$$18 = 1 \times 18$$

$$18 = 2 \times 9$$

$$18 = 3 \times 6$$

Therefore:

$$\boxed{1, 2, 3, 6, 9, 18}$$

## 3 Multiples

**Definition**

A multiple of a number is obtained by multiplying that number by an integer.

For example, multiples of 5 are:

$$5, 10, 15, 20, 25, 30, 35, \ldots$$

because:

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

$$5 \times 3 = 15$$

and so on.

### 3.1 Factors vs Multiples

| | Factors | Multiples |
|---|---|---|
| **Meaning** | Numbers that divide a number exactly | Numbers obtained by multiplying a number |
| **Example** | Factors of 12: $1, 2, 3, 4, 6, 12$ | Multiples of 12: $12, 24, 36, 48, \ldots$ |
| **Quantity** | Usually finite | Infinite |
| **Size** | $\leq$ original number | $\geq$ original number for positive integers |

## 4 Prime Factorization

**Definition**

Prime factorization means expressing a number as a product of prime numbers.

For example:

$$60 = 2 \times 30$$

$$= 2 \times 2 \times 15$$

$$= 2 \times 2 \times 3 \times 5$$

Therefore:

$$\boxed{60 = 2^2 \times 3 \times 5}$$

### 4.1 Prime Factorization Using Division

Example: Find the prime factorization of 72.

$$72 \div 2 = 36$$

$$36 \div 2 = 18$$

$$18 \div 2 = 9$$

$$9 \div 3 = 3$$

$$3 \div 3 = 1$$

Therefore:

$$72 = 2 \times 2 \times 2 \times 3 \times 3$$

$$\boxed{72 = 2^3 \times 3^2}$$

**Important Rule**

Prime factorization is one of the most important skills for solving HCF and LCM using the factor method.

# 5 Divisibility Rules

Divisibility rules help us determine quickly whether one number is divisible by another without performing complete division.

## 5.1 Divisibility by 2

A number is divisible by 2 if its last digit is:

$$0, 2, 4, 6, 8$$

Example:

$$246$$

The last digit is 6, so 246 is divisible by 2.

## 5.2 Divisibility by 3

Add all the digits.

If the sum is divisible by 3, the original number is divisible by 3.

Example:

$$123$$

$$1 + 2 + 3 = 6$$

Since 6 is divisible by 3:

$$123 \text{ is divisible by } 3.$$

## 5.3 Divisibility by 4

Look at the last two digits.

If the last two digits form a number divisible by 4, the whole number is divisible by 4.

Example:

$$316$$

Last two digits:

$$16$$

Since 16 is divisible by 4:

$$316 \text{ is divisible by } 4.$$

## 5.4 Divisibility by 5

The last digit must be:

$$0 \quad \text{or} \quad 5$$

Examples:

$$25, \quad 40, \quad 75, \quad 120$$

## 5.5 Divisibility by 6

A number must be divisible by both 2 and 3.

Example:

$$42$$

It is divisible by 2 because it is even.

Also:

$$4 + 2 = 6$$

and 6 is divisible by 3.

Therefore 42 is divisible by 6.

## 5.6 Divisibility by 8

Look at the last three digits.

Example:

$$1,216$$

Last three digits:

$$216$$

Since:

$$216 \div 8 = 27$$

the number is divisible by 8.

## 5.7 Divisibility by 9

Add all digits.

Example:

$$729$$

$$7 + 2 + 9 = 18$$

Since 18 is divisible by 9:

$$729 \text{ is divisible by } 9.$$

## 5.8 Divisibility by 10

The last digit must be 0.

Examples:

$$10, 20, 50, 100, 250$$

## 5.9 Divisibility by 11

Find the difference between the sums of alternating digits.

Example:

$$121$$

$$(1+1)-2=0$$

Since 0 is divisible by 11, 121 is divisible by 11.

## 5.10 Divisibility by 12

A number is divisible by 12 if it is divisible by both:

$$3 \quad \text{and} \quad 4$$

Example:

$$144$$

Digit sum:

$$1+4+4=9$$

So it is divisible by 3.

Last two digits:

$$44$$

44 is divisible by 4.

Therefore:

$$144 \text{ is divisible by } 12.$$

| **Number** | Divisibility Rule |
|---|---|
| **2** | Last digit is even |
| **3** | Sum of digits is divisible by 3 |
| **4** | Last two digits are divisible by 4 |
| **5** | Last digit is 0 or 5 |
| **6** | Divisible by both 2 and 3 |
| **8** | Last three digits are divisible by 8 |
| **9** | Sum of digits is divisible by 9 |
| **10** | Last digit is 0 |
| **11** | Alternating digit-sum difference is divisible by 11 |
| **12** | Divisible by both 3 and 4 |

# 6 HCF – Highest Common Factor

**Definition**

HCF stands for Highest Common Factor. It is the largest number that divides two or more numbers exactly.

HCF is also called:

$$\boxed{\text{GCD} = \text{Greatest Common Divisor}}$$

## 6.1 HCF by Listing Factors

Example: Find the HCF of 18 and 24.

Factors of 18:

$$1, 2, 3, 6, 9, 18$$

Factors of 24:

$$1, 2, 3, 4, 6, 8, 12, 24$$

Common factors:

$$1, 2, 3, 6$$

Highest common factor:

$$\boxed{6}$$

Therefore:

$$HCF(18, 24) = 6$$

## 6.2 HCF Using Prime Factorization

Example:

Find HCF of 24 and 36.

Prime factorization:

$$24 = 2^3 \times 3$$

$$36 = 2^2 \times 3^2$$

Take the common prime factors with the smallest powers:

$$2^2 \times 3$$

Therefore:

$$\boxed{HCF = 12}$$

**Important Rule**

For HCF:

$$\boxed{\text{Take common primes with minimum powers}}$$

## 7 HCF Using Euclidean Algorithm

The Euclidean algorithm is a fast method for finding HCF.

Example: Find HCF of 48 and 18.

First divide the larger number by the smaller number:

$$48 \div 18 = 2 \text{ remainder } 12$$

Now divide 18 by 12:

$$18 \div 12 = 1 \text{ remainder } 6$$

Now:

$$12 \div 6 = 2 \text{ remainder } 0$$

The last non-zero remainder is 6.

Therefore:

$$\boxed{HCF(48, 18) = 6}$$

**Shortcut**

Euclidean Algorithm:

$$\text{Larger} \div \text{Smaller}$$

Then:

$$\text{Smaller} \div \text{Remainder}$$

Continue until the remainder becomes 0.
The last non-zero remainder is the HCF.

# 8 LCM – Least Common Multiple

**Definition**

LCM stands for Least Common Multiple. It is the smallest positive number that is a multiple of all the given numbers.

Example:

Multiples of 4:

$$4, 8, 12, 16, 20, 24, \ldots$$

Multiples of 6:

$$6, 12, 18, 24, \ldots$$

The first common multiple is:

$$12$$

Therefore:

$$\boxed{LCM(4, 6) = 12}$$

## 8.1 LCM Using Prime Factorization

Example: Find LCM of 24 and 36.

Prime factorization:

$$24 = 2^3 \times 3$$

$$36 = 2^2 \times 3^2$$

For LCM, take every prime with the highest power:

$$LCM = 2^3 \times 3^2$$

$$= 8 \times 9$$

$$\boxed{LCM = 72}$$

**Important Rule**

For LCM:

$$\boxed{\text{Take every prime with maximum powers}}$$

# 9 HCF vs LCM

This is one of the most important concepts to remember.

| | HCF | LCM |
|---|---|---|
| **Meaning** | Highest Common Factor | Least Common Multiple |
| **Prime method** | Common primes only | All required primes |
| **Power** | Minimum power | Maximum power |
| **Typical use** | Grouping/dividing | Repeating/meeting |
| **Example** | Largest equal group | First common occurrence |

**Shortcut**

Remember:

$$\boxed{\text{HCF} = \text{Common} + \text{Minimum}}$$

$$\boxed{\text{LCM} = \text{All} + \text{Maximum}}$$

# 10 Relationship Between HCF and LCM

For two positive integers $a$ and $b$:

$$\boxed{HCF(a,b) \times LCM(a,b) = a \times b}$$

## 10.1 Example

Find the LCM of 12 and 18 if their HCF is 6.

Using:

$$HCF \times LCM = a \times b$$

$$6 \times LCM = 12 \times 18$$

$$LCM = \frac{12 \times 18}{6}$$

Cancel:

$$\frac{12}{6} = 2$$

Therefore:

$$LCM = 2 \times 18$$

$$\boxed{LCM = 36}$$

**Remember**

The relationship

$$HCF \times LCM = a \times b$$

is directly applicable to two positive integers.

## 11 Fast LCM Calculation

For two positive numbers:

$$\boxed{LCM = \frac{a \times b}{HCF(a, b)}}$$

Example:

Find LCM of 48 and 18.

We know:

$$HCF(48, 18) = 6$$

Therefore:

$$LCM = \frac{48 \times 18}{6}$$

Simplify first:

$$48 \div 6 = 8$$

Therefore:

$$LCM = 8 \times 18$$

$$\boxed{LCM = 144}$$

**Shortcut**

Do not always multiply first.
Cancel common factors first to reduce the calculation.
This is useful for calculation speed.

# 12 HCF and LCM of Three or More Numbers

## 12.1 HCF of Three Numbers

Example:

Find HCF of:

$$12, \quad 18, \quad 24$$

Prime factorization:

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

$$24 = 2^3 \times 3$$

Common primes:

$$2, \quad 3$$

Minimum powers:

$$2^1, \quad 3^1$$

Therefore:

$$HCF = 2 \times 3$$

$$\boxed{HCF = 6}$$

## 12.2 LCM of Three Numbers

Using the same numbers:

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

$$24 = 2^3 \times 3$$

Take maximum powers:

$$2^3 \times 3^2$$

Therefore:

$$LCM = 8 \times 9$$

$$\boxed{LCM = 72}$$

# 13 HCF and LCM of Fractions

These are useful basic formulas to know.

For positive fractions:

$$\boxed{HCF\left(\frac{a}{b}, \frac{c}{d}\right) = \frac{HCF(a,c)}{LCM(b,d)}}$$

and:

$$\boxed{LCM\left(\frac{a}{b}, \frac{c}{d}\right) = \frac{LCM(a,c)}{HCF(b,d)}}$$

## 13.1 Example

Find the LCM of:

$$\frac{2}{3}, \quad \frac{4}{5}$$

LCM of numerators:

$$LCM(2,4) = 4$$

HCF of denominators:

$$HCF(3,5) = 1$$

Therefore:

$$LCM = \frac{4}{1}$$

$$\boxed{LCM = 4}$$

**Remember**

Fraction HCF/LCM formulas are useful, but for your quick revision goal they are lower priority than factors, prime factorization, HCF, LCM and divisibility.

# 14 Using HCF for Simplification

HCF is extremely useful for reducing fractions.

Example:

$$\frac{24}{36}$$

Find HCF:

$$HCF(24, 36) = 12$$

Divide numerator and denominator by 12:

$$\frac{24 \div 12}{36 \div 12}$$

$$= \frac{2}{3}$$

Therefore:

$$\boxed{\frac{24}{36} = \frac{2}{3}}$$

## 14.1 Fast Cancellation

Consider:

$$\frac{24 \times 35}{18}$$

Instead of calculating $24 \times 35$ first, simplify:

$$\frac{24}{18} = \frac{4}{3}$$

Therefore:

$$\frac{24 \times 35}{18} = \frac{4 \times 35}{3}$$

$$= \frac{140}{3}$$

The main idea is:

**Cancel common factors before multiplying.**

# 15 Basic Word Problems

## 15.1 When to Use HCF

HCF is commonly used when the question involves:

- largest possible equal groups,
- largest equal size,
- dividing things equally,
- no remainder,
- maximum length/size that divides all quantities.

**Example**

You have 24 apples and 36 oranges. You want to make the largest possible number of identical groups without anything left.
The number of groups must divide both 24 and 36.
Therefore, use HCF:

$$HCF(24, 36) = 12$$

So the largest possible number of groups is:

$$\boxed{12}$$

## 15.2 When to Use LCM

LCM is commonly used when the question involves:

- repeating events,
- things happening together,
- bells ringing together,

- schedules repeating,
- finding the first common occurrence.

**Example**

A bell rings every 6 minutes and another bell rings every 8 minutes.
When will they ring together again?
We need the first common multiple:

$$LCM(6, 8) = 24$$

Therefore:

$$\boxed{24 \text{ minutes}}$$

**Shortcut**

Think:

$$\boxed{\text{Grouping / Dividing} \rightarrow HCF}$$

$$\boxed{\text{Repeating / Together} \rightarrow LCM}$$

# 16 Mental Calculation Skills

If your goal is to become faster at mathematics, learning HCF and LCM alone is not enough. You should also build basic number recognition.

## 16.1 Multiplication Tables

Know multiplication tables from:

$$1 \text{ to } 20$$

At minimum, be very comfortable with:

$$11, 12, 13, 14, 15, 16, 17, 18, 19, 20$$

## 16.2 Squares

Memorize squares from:

$$1^2 \text{ to } 30^2$$

Some important values:

$$10^2 = 100$$

$$12^2 = 144$$

$$15^2 = 225$$

$$20^2 = 400$$

$$25^2 = 625$$

$$30^2 = 900$$

## 16.3 Cubes

Learn cubes at least from:

$$1^3 \text{ to } 15^3$$

Examples:

$$2^3 = 8$$

$$3^3 = 27$$

$$4^3 = 64$$

$$5^3 = 125$$

$$10^3 = 1000$$

## 16.4 Recognizing Common Factorizations

Try to recognize these immediately:

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

$$24 = 2^3 \times 3$$

$$30 = 2 \times 3 \times 5$$

$$36 = 2^2 \times 3^2$$

$$48 = 2^4 \times 3$$

$$60 = 2^2 \times 3 \times 5$$

$$72 = 2^3 \times 3^2$$

$$84 = 2^2 \times 3 \times 7$$

$$90 = 2 \times 3^2 \times 5$$

$$120 = 2^3 \times 3 \times 5$$

$$144 = 2^4 \times 3^2$$

$$180 = 2^2 \times 3^2 \times 5$$

$$240 = 2^4 \times 3 \times 5$$

**Shortcut**

The faster you recognize prime factors, the faster you can calculate HCF and LCM mentally.

# 17 Common Mistakes

## 17.1 Mistake 1: Confusing Factor and Multiple

For 6:

Factors:

$$1, 2, 3, 6$$

Multiples:

$$6, 12, 18, 24, \ldots$$

They are not the same thing.

## 17.2 Mistake 2: Using Maximum Powers for HCF

Incorrect:

$$HCF(12, 18)$$

using maximum powers.

Correct:

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

HCF uses minimum powers:

$$2^1 \times 3^1 = 6$$

## 17.3 Mistake 3: Using Minimum Powers for LCM

LCM requires maximum powers.

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

Therefore:

$$LCM = 2^2 \times 3^2 = 36$$

## 17.4 Mistake 4: Multiplying First

For:

$$\frac{48 \times 18}{6}$$

Do not calculate:

$$48 \times 18 = 864$$

first.

Instead:

$$48 \div 6 = 8$$

then:

$$8 \times 18 = 144$$

This is faster and reduces errors.

# 18 One-Page Memory Sheet

**Factor**
A number that divides another number exactly.

**Multiple**
A number obtained by multiplying a number by an integer.

**Prime Factorization**
Express a number as a product of prime numbers.

**HCF**

$$\boxed{\text{Common primes} + \text{Minimum powers}}$$

**LCM**

$$\boxed{\text{All required primes} + \text{Maximum powers}}$$

**HCF–LCM Relationship**

$$\boxed{HCF \times LCM = a \times b}$$

**Fast LCM**

$$\boxed{LCM = \frac{a \times b}{HCF}}$$

**Fast HCF**
Use the Euclidean algorithm.

**Word Problems**

$$\boxed{\text{Grouping / Largest equal size} \rightarrow HCF}$$

$$\boxed{\text{Repeating / Together again} \rightarrow LCM}$$

# 19 Practice Questions

## 19.1 Level 1 – Basic

1. Find all factors of 24.
2. Find all factors of 36.
3. Write the first 10 multiples of 7.
4. Find the prime factorization of 48.
5. Find the prime factorization of 90.

6. Is 97 prime or composite?
7. Is 121 divisible by 11?
8. Is 324 divisible by 3?

## 19.2 Level 2 – HCF

1. Find HCF of 12 and 18.
2. Find HCF of 24 and 36.
3. Find HCF of 48 and 60.
4. Find HCF of 72 and 90.
5. Find HCF of 84 and 126.
6. Find HCF of $24, 36, 48$.
7. Find HCF of $60, 90, 120$.

## 19.3 Level 3 – LCM

1. Find LCM of 4 and 6.
2. Find LCM of 8 and 12.
3. Find LCM of 12 and 18.
4. Find LCM of 24 and 36.
5. Find LCM of 15 and 20.
6. Find LCM of $12, 18, 24$.
7. Find LCM of $8, 12, 20$.

## 19.4 Level 4 – Mixed Calculation

1. Find both HCF and LCM of 18 and 24.
2. Find both HCF and LCM of 36 and 48.
3. If HCF of two numbers is 6 and their LCM is 72, find their product.
4. If HCF is 8, LCM is 120, and one number is 24, find the other number.
5. Simplify $\frac{36}{48}$ using HCF.
6. Simplify $\frac{84}{126}$ using HCF.

### 19.5 Level 5 – Practical

1. One bell rings every 6 minutes and another every 8 minutes. When will they ring together?
2. Three machines complete a cycle every 4, 6 and 8 minutes. When will all three complete a cycle together?
3. You have 24 red balls and 36 blue balls. What is the maximum number of identical groups you can make without anything remaining?
4. A rectangular floor is 24 m long and 36 m wide. What is the largest square tile that can cover the floor exactly?

## 20 Quick Revision Plan

For your goal of revising mathematics quickly, use the following three-day plan.

### Day 1 – Number Foundation

Study:

- Factors
- Multiples
- Prime and composite numbers
- Prime numbers up to 100
- Divisibility rules
- Prime factorization

Practice for:

45–60 minutes

### Day 2 – HCF and LCM

Study:

- HCF by factors
- HCF by prime factorization
- Euclidean algorithm
- LCM by multiples

- LCM by prime factorization
- HCF–LCM relationship
- Fast LCM calculation

Practice for:

45–60 minutes

## Day 3 – Application and Speed

Study:

- HCF/LCM of three or more numbers
- Fractions
- Simplification
- Cancellation
- Word problems
- Mental calculation

Practice for:

45–60 minutes

## After Revision

After completing the topic, do not repeatedly reread the theory.
Instead, spend:

10–15 minutes

every few days solving mixed questions.
The goal is to make these operations automatic:

Factorization

↓

HCF / LCM

↓

Simplification

↓

Fast Calculation

## 21 Final Summary

The complete concept map for this topic is:

**Numbers**
↓
**Factors and Multiples**
↓
**Prime Numbers**
↓
**Prime Factorization**
↓
**Divisibility Rules**
↓
**HCF**
↓
**LCM**
↓
**HCF–LCM Relationship**
↓
**Simplification and Calculation**
↓
**Practical Word Problems**

The most important rules to remember are:

$$\boxed{\text{HCF} = \text{Common} + \text{Minimum Powers}}$$

$$\boxed{\text{LCM} = \text{All} + \text{Maximum Powers}}$$

$$\boxed{HCF \times LCM = a \times b}$$

$$\boxed{LCM = \frac{a \times b}{HCF}}$$

$$\boxed{\text{Grouping} \rightarrow HCF}$$

$$\boxed{\text{Repeating} \rightarrow LCM}$$

**Main Goal**

Do not try to memorize dozens of tricks.
Build strong number sense, become comfortable with prime factorization, memorize important multiplication tables and squares, and practice small calculations repeatedly.
This will make HCF, LCM and many other areas of mathematics faster.